MPG No. 23.3.84 $6'/m$ [Type III, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & 0 & 0 & -M_{xxzx} \\ M_{xxzx} & -M_{xxzx} & 0 & 0 & 0 & M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4s}^{(1)}$$
$$M_{xxzx} = \mathbb{M}_{4c}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & 0 & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & 0 & 0 & -M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{T}_{3c}^{(1)}$$
$$M_{xxzx} = -2\mathbb{T}_{3s}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & M_{xxzx} & 0 \\ -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxzx} & -M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{T}_{3c}^{(2)}$$
$$M_{xxzx} = \mathbb{T}_{3s}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$$
$$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 & M_{xxzy} & M_{xxxz} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 & -M_{xxzy} & -M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & -M_{xxxz} & M_{xxzy} & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} + \mathbb{T}_{3c}^{(2)}$$
$$M_{xxzx} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} + \mathbb{T}_{3s}^{(2)}$$
$$M_{xxzy} = \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} - \mathbb{T}_{3c}^{(2)}$$
$$M_{xxxz} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} - \mathbb{T}_{3s}^{(2)}$$
$$M_{yzxx} = \mathbb{M}_{4s}^{(1)} - 2\mathbb{T}_{3c}^{(1)}$$
$$M_{yzxy} = -\mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$$
$$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & M_{yyyz} \\ 0 & 0 & 0 \\ -M_{xxxz} & M_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyz} \\ 0 & 0 & 0 \\ M_{xxxz} & -M_{yyyz} & 0 \\ 0 & 0 & -M_{xxxz} \\ -M_{yyyz} & -M_{xxxz} & 0 \end{bmatrix}$$

$$M_{xxxz} = \mathbb{M}_{4c}^{(1)}$$
$$M_{yyyz} = -\mathbb{M}_{4s}^{(1)}$$